

CPSC 3660 – Winter 2026

Introduction to Database Systems

Course Project

Your task is to build a *real* (though small-scaled) business database application. The project reinforces the concepts discussed in the class.

(A) Project Group

Students can form a group of *at most five* (5) members. Each group needs to select a member as the group leader, who is responsible for submitting your project (as shown below). You are responsible for finding your partners. All the members in a group will be awarded the same marks.

(B) Project Description

Lethbridge JonesAuto

(B.1) Background

Lethbridge JonesAuto is a local used-car dealer. As the volume of business data it needs to handle becomes larger and larger, it urgently needs a database to track the cars purchased and sold.

In JonesAuto, cars are bought at auctions. The company pays cash at the auctions and brings the cars back for cleaning up. Most of the times the cars need minor repairs before they can be sold to customers. When cars are being sold, the company generally finances the sale. Customers rarely have good credit. Therefore, the company prices the cars such that the customers cover the original purchase cost of the car with a down payment. The rest of the purchase prices are financed. The purpose of doing this is that, even if a customer stopped paying the loan, the company does not lose any money. Consequently, a key aspect of the company's profitability is to find cars at lower prices.

Currently, JonesAuto mainly conducts several business activities, mainly including car purchases, car sales, car warrants and payments. At this moment, all the operations are done manually on papers. We discuss each one of them in turn in detail in the following.

(B.2) Business Activities

Car Purchases

Lethbridge JonesAuto has a group of *buyers* who travel the nearby areas to attend various auctions. They look for good deals. To get the good prices, the company is willing to pick up the cars other companies do not want. In particular, the cars often need minor repairs. These repairs and costs are estimated at the time of the purchase by the buyer, based on their previous experiences. Then the actual costs are recorded as the repair work is done.

Purchase						
Date		Seller/Dealer				
Location						
Auction (Yes/No)						
Vehicle Info			#	Problem	Est. Repair cost	Actual cost
Make	Model	Year				
Color						
Miles						
Condition						
Book Price						
Price Paid						

Form 1. Purchase

The repairs usually involve simple body work to fix scratches, body dents, broken windows, or cosmetic work on the interior. The company has a policy: it does not purchase cars that need major repairs. As a result, all work is able to be performed at a small shop at low prices.

Form 1 shows the basic form the buyers will fill out when a car is purchased. The individual problems spotted are numbered, but not in any particular order. Remember that the buyer is looking at dozens of cars at a time and needs to quickly record basic data about the cars during an initial inspection before the bidding begins. The book price is the average price for the vehicle as listed in the industry guidance books. At some point, the buyers would like to have a portable version of the database so they can fill out this form and have it calculated totals while at an auction, which means that we need an IE-like browser to access our database from anywhere through the Internet.

Car Sales

Sales of the cars at Lethbridge JonesAuto are relatively traditional. A *salesperson* helps a customer select a car and negotiates the price. When the customer decides to proceed, the salesperson needs to fill out the information in Form 2.

Sale		Salesperson					
Date		Last Name					
Total due		First Name					
Down payment		Phone					
Financed Amount		Commission					
Customer		Employment History					
Customer ID		Employer	Title	Super.	Phone	Address	Start
Phone							
Last Name							
First Name							
Address							
City							
State							
Zip							
Vehicle Info							
Miles							
Condition							
Book Price							

Sale Price Style (Sedan, SUV, ...) Interior Color	
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Form 2. Sales

The employee commission is usually added to the form after the customer signs the sale. But sometimes an employee lists a lower-than-normal commission to help sell the car. Since JonesAuto finances a portion of the price, customers are asked to list their current and last few employers – in case JonesAuto needs to repossess the vehicle.

Warranties and Payments

JonesAuto also offers a variety of warranties as a major promotional strategy. The warranties help sell cars because customers believe that a warranty lowers their risks. Yet, because the cars are relatively sound to begin with, few warranties arise. So the company makes a solid profit on the warranties. Consequently, a *salesperson* is encouraged to sell warranties and receives a relatively high 25 percent commission on them.

As shown in Form 3, separate warranties are offered for various components. For instance, an exterior warranty covers the paint, while a drive-train warranty covers the various engine and transmission elements.

Note that each warranty only covers the listed items. Customers are given several options, including the length of warranty and the level of the deductible. Of course, longer warranties and lower deductibles cost more. Their values are specified in the warranty policies. But the salesperson needs to enter them on this form so everyone knows exactly which warranty was purchased. The customer also has the option of paying the warranty fees upfront as a lump sum. But most of them decide to finance them over time in the form of monthly payment. These values are entered by the salesperson using a predefined chart – essentially the company charges one percent per month interest rate for financing the warranty costs.

Vehicle Info			Employee (Salesperson)	
Customer			Last Name	
Warranty Sale Date			First Name	
Total Cost			Phone	
Monthly Cost				
Warranty	Start Date	Length	Cost	Deductible
Items covered				
More warranties				

Form 3. Warranties

Payments

To provide documentations for situations where complaints arise, or when JonesAuto has decided to repossess a car, the company records all of the payments made by customers.

Customer				
Customer ID				
Gender				
Data of Birth				
Number of later payments				
Average number of days late				
Payment Date	Due	Paid Date	Amount	Bank Account

Form 4. Payments

Form 4 shows the basic form. To provide faster lookups (on the paper forms), the clerk also updates the number of late payments and the average number of days late whenever a payment is late. JonesAuto uses date of birth, gender, and tax payer ID to track the credit report on a customer. Really bad customers are also reported to the credit bureau.

(B.3) Summary

As the number of the transactions booms, Lethbridge JonesAuto urgently needs a DBMS to manage its business activities. On average, the company sells 10 to 20 cars a week and generally tries to keep no more than 50 cars on the lot at one time, which not only provides a variety of choices to customers but also makes the parking management easier. However, if good deals pop up at auctions, the buyers are encouraged to purchase them. But they need to hold off on buying the marginal cars.

Your task is to design and develop a DBMS for Lethbridge JonesAuto.

(C) Project Stages and Submission

Stage 1 – Overall design

Read carefully the story of Lethbridge JonesAuto. Draft your initial proposal for the database management system. You do not need to submit the draft. It is just for your own record and is used for your group discussions.

In the proposal draft, roughly identify the system goals, the main forms and reports of the system, and possible users of the system. You may also want to discuss some other issues that are not considered by JonesAuto at this moment.

Create a list of all the forms and reports that JonesAuto might use. Explain the purpose of each form or report using your own word.

Stage 2 – ER diagram design

Discuss the business activities in detail among your group members and create a list of tables for each form and report.

Draw the corresponding ER diagram and create the database using PHP or some MySQL script. Populate your database with some simulated (sample) data, assuming that we need data to test our initial design.

Stage 3 – Forms and reports design

Build and test at least *three* initial input forms and *three* initial output reports, according to your design (to at least support the above discussed business activities). Improve these forms and reports to make them easier to use. Test your forms and reports with sample data.

Stage 4 – Query design

Build additional forms and reports, in addition to those in Stage 3. Improve and test all of them.

Create at least 10 queries. At least 5 of them are *advanced* queries in the sense that they involve more than one table and/or need sub queries. For each query, list the corresponding business question you try to answer.

Stage 5 – Final submission

- (1) Create a small tutorial as how to use your system.
- (2) Write a report for your project. Clearly state your development stages and in each stage, what you have done, including ER-diagram, input forms and output reports, and etc. You can claim any assumptions behind your design and give reasons for them. Also state what you have gained during this project and what improvement you could do in the future. You may want to include other parts to support your project efforts.

You also need to state the names and student IDs of your group on the front page of your report.

- (3) Submit your PHP and HTML code, in a *.zip file, by the leader of your group on the Moodle.
- (4) Do a demo to show that your project works.

There might be additional submission of the testing data in the database. If this is the case, it will be announced to the whole class.

For (1) and (2), please submit a hard copy of both of them. Your project will be examined against them.

The details of the due date and demo date of the project will be announced on the Moodle in the middle of March.

(D) Additional Comments

To truly learn how to build business database applications, you need to work on a real project and get yourself on-site experiences. The project we introduce here is generally simpler than real-life applications. But it is still complex enough to illustrate possible similar problems you might encounter in the future.

One of the most challenging aspects in doing a database project is that it will definitely contain a level of uncertainty and ambiguity. When you start a real-life project, you never know exactly what the project is going to proceed and involve. You encounter contradictions, uncertainty, and confusion over terms and goals.

In a real environment, you can then talk with ordinary users, on-site technicians, managers, and etc. on these problems. Gradually, you will get more experiences. Sometimes this process needs several rounds of communications.

Unfortunately, for a course project, we do not have such an environment for you. Though I can answer some questions, you need to make your own decisions and interpretations.

When you start to analyze the individual input forms and output reports, you need to figure out the overall purpose of each of them. You should be able to summarize such a purpose in a short paragraph (containing one or two sentences) using your own word. Keep a list of these forms so that you always remember the overall purpose of the application.

It is a good strategy to start with small tables and add more tables and features slowly. Start with an initial set of tables and keys that you are certain are correct. Then begin building forms. Add more attributes and tables later on as you need. If you initial tables and keys are correct, you should be able to add new columns and tables without altering the existing work too much.

Normalization is a process that you might need to rework for your project before it is complete. Try to develop a list of good normalized tables before you begin creating forms and reports, but leave yourself enough time to go back and change the tables if you spot problems.

The design and development of a database application is a painful or at least a struggling process. So the most important advice at this moment is that you start working on the project as soon as possible. Do not put the project off until the end of the semester.

One final word of advice is that for the course project, it is usually better to complete half a project where everything works, instead of a large and messy one where nothing works.